RIZKI RAMADHANI

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EDUCATION

IPB University (Institut Pertanian Bogor) Bogor, Indonesia
Applied Bachelor (D4) in Computer Engineering Technology • ***GPA: 3.77 / 4.00*** Aug 2024 – Present (Exp. 2028)

- **College of Vocational Studies:** Focused on hardware-software co-design, real-time telemetry systems, and network engineering.
- **Relevant Coursework:** Embedded Systems, Microcontroller Architecture, Computer Networks, Database Systems, Object-Oriented Programming, Data Structures & Algorithms.

TECHNICAL SKILLS

Languages: C/C++, TypeScript, JavaScript, Python, PHP, SQL (PostgreSQL/MySQL), HTML5, CSS3

IoT & Embedded: ESP32, Arduino (Nano, Mega), FreeRTOS basics, Sensor Interfacing (DHT22, Soil Moisture), REST/HTTP Telemetry, MQTT

Backend & ML: AdonisJS, Laravel, Node.js, FastAPI, Prisma ORM, Eloquent ORM, scikit-learn (time-series forecasting)

Frontend & UI: React.js, Vite, Ionic Framework, Livewire, Tailwind CSS, Blade Templates

Developer Tools: PostgreSQL, MySQL, Docker, Git/GitHub, Linux/Bash, Postman, Vercel

TECHNICAL PROJECTS

MUSH: End-to-End IoT Monitoring & Forecasting Ecosystem github.com/rizkiramadhani-001/mush-server
Full-Stack & Embedded Architecture • AdonisJS, React, ESP32, FastAPI, PostgreSQL 2026

- **Telemetry Ingest & REST API (mush-server):** Architected modular AdonisJS backend with Prisma ORM across 15 REST endpoints; added composite indexes on PostgreSQL time-series tables to achieve query latencies under 80ms.
- **Interactive Client Dashboard (mush-react):** Built cross-platform monitoring application using Ionic React and Vite, featuring historical sensor visualization and dynamic threshold alerts for ambient and substrate metrics.
- **Time-Series Forecasting Service (mush-ml):** Developed microservice using Python, FastAPI, and scikit-learn to generate environmental predictions from ingested logs, completing inference requests in <150ms.
- **Responsive Landing Page (landing):** Created dedicated showcase and product documentation page using semantic HTML5, Tailwind CSS, and vanilla JavaScript.

MyKeuangan: Personal Financial Management System github.com/rizkiramadhani-001/MyKeuangan
Web Application • Laravel, Livewire, Tailwind CSS, MySQL, Vite 2025 – 2026

- Engineered reactive financial ledger with real-time income and expense tracking, category filtering, and balance calculations powered by Laravel Livewire components.
- Streamlined database queries with Eloquent ORM relationships, eliminating redundant lookups and generating comprehensive monthly cashflow reports.

Multi-Node Sensor Firmware Architecture github.com/rizkiramadhani-001/Arduino-Project-4-Node-and-Mega
Firmware Developer • C++, Arduino IDE, Arduino Mega & Nano 2025 – 2026

- Developed modular C++ firmware for distributed sensor nodes using Arduino Nano and central coordinator routines on Arduino Mega.
- Implemented serial communication buffers and sensor calibration logic to ensure stable multi-sensor data sampling without memory leaks.

Sensorix Team Portfolio sensorix-porto.vercel.app
Frontend Contributor • JavaScript, HTML5, CSS3, Vercel 2026

- Collaborated on building a responsive project portfolio for the Sensorix engineering team, deployed and continuously hosted on Vercel.